

Morgan Patel

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EDUCATION

Ridgeway University | Ph.D. candidate in Statistics | Metro City, USA | Sep 2021 - Expected May 2026

Ridgeway University | B.S. in Mathematics | Metro City, USA | Sep 2016 - May 2020

EXPERIENCE

Northpeak Capital | Quantitative Research Intern | Metro City, USA | Jun 2025 - Aug 2025

- Built a short-horizon order-book imbalance signal for liquid index futures that added 0.4 Sharpe to the desk's book over 18 months of out-of-sample backtest after costs.
- Cut the signal's daily turnover from 34% to 21% with a cost-aware position smoother, keeping 90% of gross returns and lowering estimated slippage by a third.
- Validated the signal on 6 years of tick data, using a Sharpe-ratio test for heteroskedasticity to confirm there was no look-ahead leakage.
- Joining 120 microstructure features point-in-time across six venues, deduplicating late prints and versioning each schema, built a feature store the team reused in two later projects.
- Tuned the signal's parameters on the full 6-year sample, then reported the Sharpe of a walk-forward test over the same period as out-of-sample.
- Documented the backtest assumptions, transaction-cost model and known failure regimes in the desk's research wiki for future interns.

Ridgeway University | Graduate Research Assistant, Statistical Learning Lab | Metro City, USA | Jun 2021 - Aug 2021

- Built the lab's simulation pipeline on the shared cluster, cutting a 2,000-run Monte Carlo study from 3 days to 5 hours and making runs reproducible by seed.
- Derived a variance bound for a sparse regression estimator; my proof tightens the previous bound by a log factor and is now Section 3 of a paper under review at JASA.
- Taught weekly recitations for 60 students in graduate probability, writing 12 problem sets and earning a 4.8/5 teaching rating.
- Released an open-source R package for high-dimensional covariance estimation while maintaining the lab's shared cluster, organizing the weekly reading group and grading for two courses, which was downloaded 3,000 times in its first year.

Sunrise Bakery | Assistant Store Manager | Metro City, USA | Jun 2020 - May 2021

- Managed opening shifts and a team of 6 bakers and cashiers, keeping the store within its weekly labour budget.
- Ran daily stock counts and supplier orders, cutting unsold bread from 12% to 7% of production.

PROJECTS

Volatility Forecasting Study | Independent Research | Python, PyTorch | Jan 2024 - Present

- Beat a HAR-RV baseline's out-of-sample QLIKE loss by 7% on 30 equity indices with a temporal convolutional model trained on realized-volatility features.
- Cut forecast error from 0.20 to 0.15, a 33% improvement, by adding realized-volatility features and an asymmetric loss.
- Improved the model's directional hit rate by 6% on 30 equity indices, from 52% to 58%, with a temporal convolutional model.

Kaggle Market Prediction Competition | Team of 3 | Python | Mar 2023 - Jun 2023

- Placed 41st of 2,900 teams with a gradient-boosting ensemble on 300 engineered features, finishing in the top 2% of the private leaderboard.
- Cut validation leakage by switching to time-grouped folds, which closed a 0.02 gap between local validation and leaderboard scores.
- Wrote the team's feature-selection script using permutation importance, trimming 900 candidate features to 300 without losing validation score.

SKILLS

Programming: Python, R, SQL, C++, Kafka

Methods: time-series econometrics, Bayesian inference, gradient boosting, PyTorch